

Machine Data Connector (tmMDC)

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Revision History

Version	Date	Author	Changes made
1.1	March 24, 2023	Mohammad Shahid	Added Remote Control feature. Updated ' About tmc ' section.
1.0	June 24, 2022	Tanmay Mallick	Creation

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1 About tmc

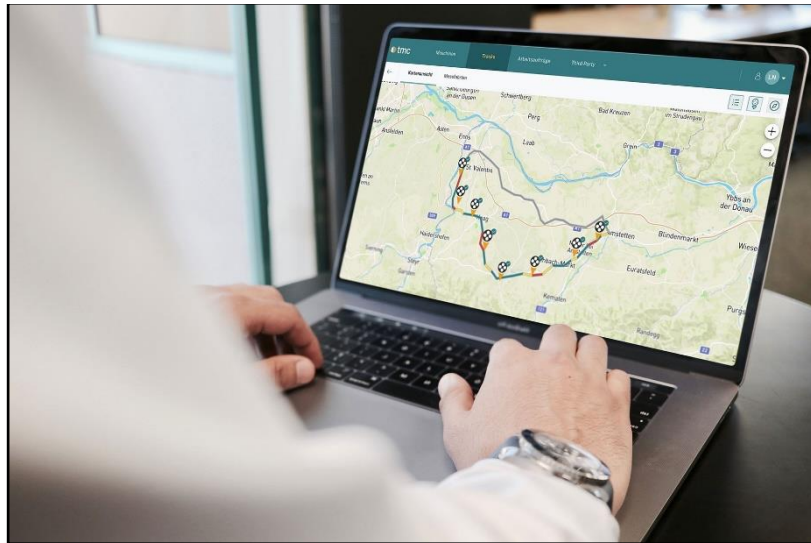


Figure 1. Digital track maintenance - System solutions for maximum efficiency and availability in rail-infrastructure maintenance

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Founded in Austria, headquartered in Linz, and efficiently backed by its subsidiary Vienna and Delhi offices, tmc employs over 100 of the world's brightest scientists, developers, project managers and railway digitalization enthusiasts. An established technological force, tmc covers customer needs across Europe, North America, Japan, and Australia.

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2 About this Manual

Title: Machine Data Connector (tmMDC) User Manual

Category: Fleet

Product: tmMDC

Applicable product release version: 4.2.0

Required release date: 24.03.2023

Owner: Matthias Lehner

2.1 Intended Purpose

The main goal of this manual is to assist the end-users by providing them with clear and comprehensible information about the tmMDC application running on the tamping machines.

2.2 Target Audience

This manual is intended for maintenance personnel who must maintain the tmMDC product. The technician should be familiar with data exchange protocols.

2.3 Document Conventions

In this document, certain words or phrases can be represented in different fonts or styles. This highlighting is systematic, with following predefined conventions:

Convention	Description
<i>Italic font style</i>	Italicized text signalizes <i>newly introduced terms</i> or <i>gently emphasized words</i> .
Bold font style	This style is for strongly emphasized words .
<u>bold underline font style</u>	Use this style for referencing to other sections, tables or figures. References also have to be linked to the corresponding source. <i>Example:</i> For more information, see <u>Figure 1</u> .
Courier New	Text written in <code>Courier New</code> signalizes file paths, commands, or an expected output from a console. <i>Example:</i> The file can be found in <code>C:/Windows/recordings/</code>
[KEY]	This style signalizes the action to press a [KEY], a [SEQUENCE] -> [OF] -> [KEYS] or a [KEY] - [COMBINATIONS]. <i>Example:</i> Press [ALT] - [A] to activate an overlay
<Replaceable>	All sections in this style are meant to be replaced by the user. Can also be in <code><Courier New></code> when part of, e.g. a command.
DEVICE NAME	This style is defined for device names represented in ALL CAPS. For example, COM1.
UI Terms	This style is for all User Interface terms represented as <Initial Caps + Bold >. Link multiple steps of navigation with >. <i>Examples:</i> Import/Export dialog box, Options list. Select File > Load File to open a separate dialog.

2.3.1 Box-shaped Highlights

Additionally, this document includes box-shaped areas, highlighting relevant information to the reader with several levels of importance:



Note: This box signals important information, e.g. memory limitations, user scenarios...



Tip: Here you can find additional information such as alternate navigation, shortcuts, and techniques...



Caution: This one is signaling a precaution to be taken to avoid minor/moderate injury or something that can be problematic for a smooth operation of the application...



Warning: Highlighting an important action to be avoided, which could potentially result in physical harm to user or hardware or can inflict serious issues to the application or its intended workflow, like loss of data or long downtimes of the operability...

3 Introduction

The Machine Data Connector (**tmMDC**) is a data logging system. This is developed to collect and record real time data from various machines (for example, Tamping Machines, Rail Cleaning Machines, and Lifting Machines). These data packages are collected, prepared, encrypted, and transferred to a secure online data storage in the cloud. Specifically, it delivers live machine data such as **machine data** or **measurement data** to a database to display it on Machine Condition Observer (**tmMCO**). It is a stand-alone system, which manages all the required part systems internally.

The tmMDC also has a unique low-power mode. If the machine is switched off, as is often the case during transfers, the tmMDC still measures the machine temperature, the voltage of the machine's battery and it sends its current physical location.

If there is no Global System for Mobile communication (GSM) connection, it can buffer the collected data for up to three months. It is the backbone of the networked machine and is integrated flawlessly with other on-board and off-board machine products.

3.1 Fields of Application

The tmMDC is used all over the world, it collects the machine data and sends to tmMCO. Fleet managers from Europe, USA, Japan, and India rely on the tmMDC.

4 Installation Guide

Please refer to **tmMDC - Installation Guide**, this provides a detailed information about how to install, update, uninstall, and configure the tmMDC application on a Windows operating system.

5 Getting Started

This section will explain the available system-wide parameters and configuration options on the tmMDC. This includes settings for the following items: General tmMDC system settings, tmSync, DGPS, and VPN Connectivity.

The four main parts of the tmMDC system are:

- **Main tmMDC System**
 - Please refer to **tmMDC DataSheet**. This document provides detailed information about the tmMDC interfaces, how to use, and the other related information on Data Storage etc.
- tmSync (For detail information refer to **tmSync - User Manual**).
- Differential Global Positioning System (DGPS).
- VPN Connection.

6 VPN Connection

Latest tmMDC firmware (4.0 or higher) supports VPN connection to the machine network via tmMDC using OpenVPN community edition.

6.1 Preconditions

The OpenVPN server can manage the VPN connectivity to a given tmMDC device. Before establishing the VPN connection with tmMDC device, do the following:

1. OpenVPN community edition 2.5.X must be installed in user PC.



Note: OpenVPN 3.X not supported.



Tip: Download the OpenVPN community software from the URL: <https://openvpn.net/community-downloads/> (as example, OpenVPN-2.5.4-I604-amd64).

2. tmMDC Firmware version 4.0 or higher.
3. tmMDC coprocessor version 4.0.6 or higher
4. With the installation of firmware, new configurations will also be installed. VPN is enabled by default. If not enabled, you need to send a configuration update via tmMCO registry.
5. In tmMDC two factor authentication must be turned on.

6.2 Machine Remote Control

The Machine Remote Function can be configured via tmMDC configuration update. This function can only be enabled when the tmMDC is in low-power mode.

If the tmMDC is required to be activated through a command request, digital output 2 (pin 13) can be used to send a wake-up pulse to the system.

The digital pulse will be sent once with the below parameters:

- Voltage: 24 V
- Duration: 200 ms

6.3 Usage of VPN Tunnel via tmMCO

VPN can be used to connect to the Machine network from anywhere in the world Via tmMDC.

1. Login to tmMCO portal with your login credentials.
<https://m-c-o.com/login>



Figure 2. Login to tmMCO

2. Click on the **VPN TUNNEL**.

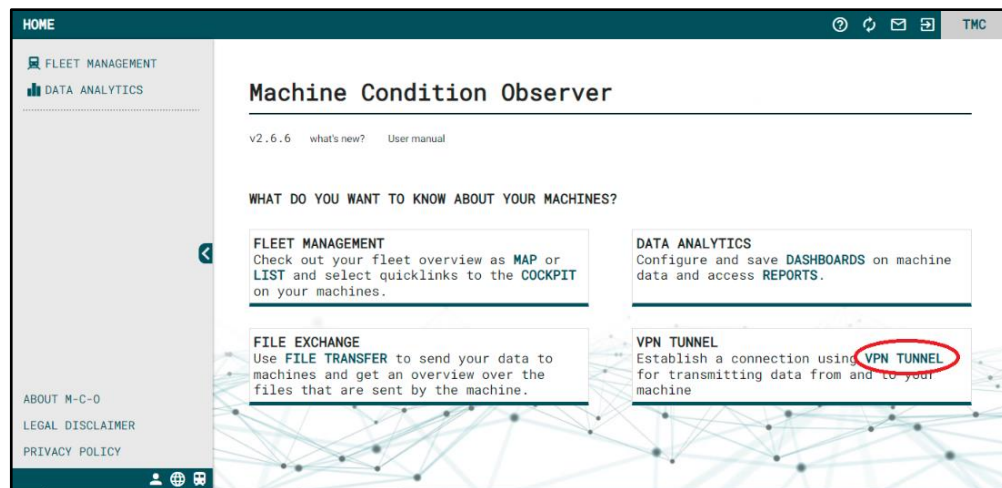


Figure 3. Selection of VPN TUNNEL

6.3.1 Select the Machine and Start Tunnel

1. Select the machine from the available list of configured machines.
2. Click on **Start Connection**.

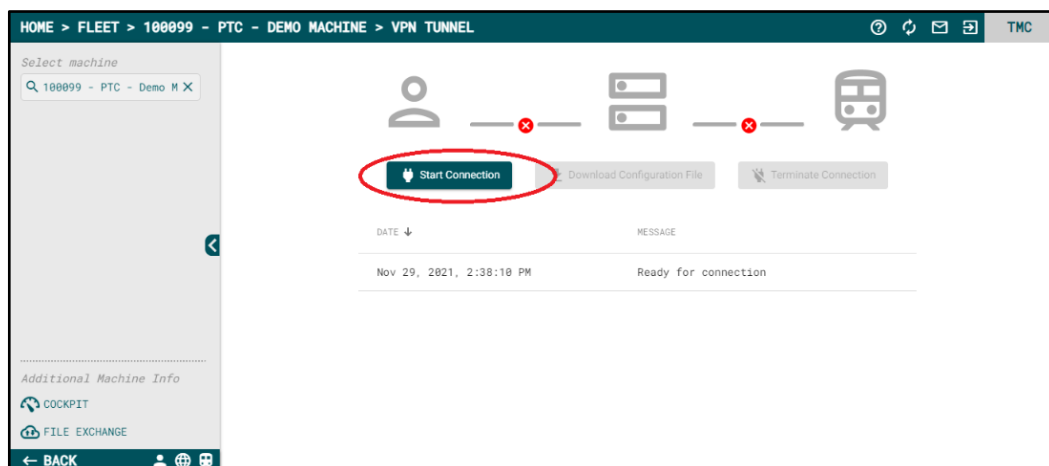


Figure 4. Select 'Start Connection' from the VPN Tunnel Page

6.3.2 Download User Configuration

1. Wait for the tmMDC to be connected.
2. **MDC Connected** message will appear in the display.
3. **"Download Configuration File"** will be active as shown in **Figure 5**.
4. Click on **"Download configuration file"** to download the config file.

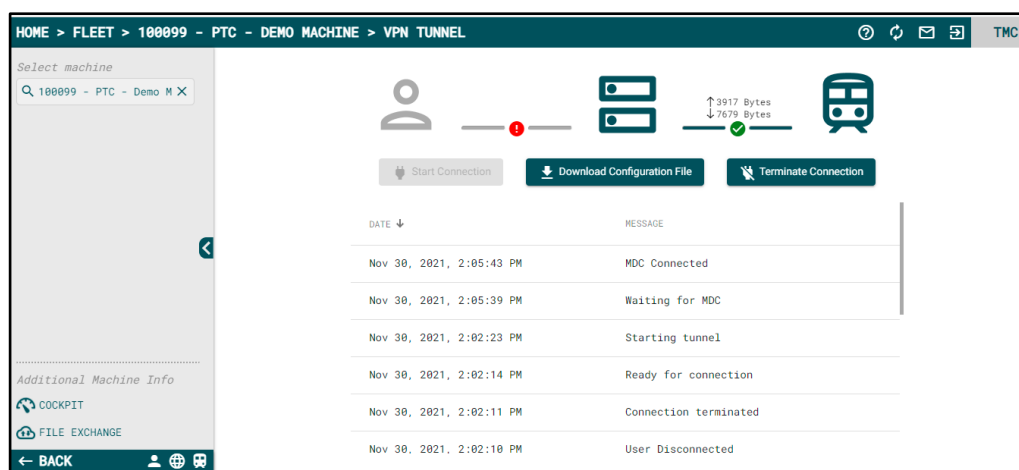


Figure 5. Download Configuration File

6.3.3 Windows Connection

1. OpenVPN GUI instance can be seen running on the task bar.

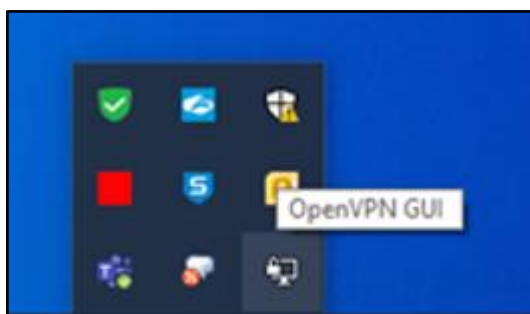


Figure 6. OpenVPN pop-up

2. Right-click on the OpenVPN instance import the downloaded configuration file.

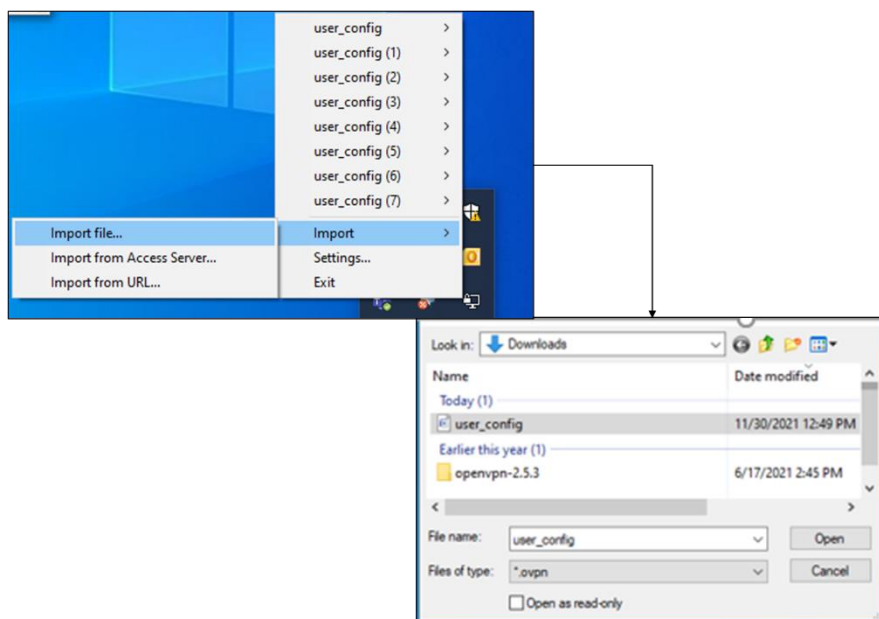


Figure 7. Import the Config File

3. A message appears on the screen after successfully importing the downloaded file.

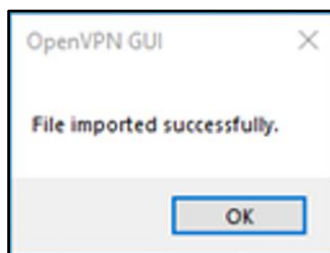


Figure 8. File Imported Successfully Message Box

4. Connect using the imported configuration file.

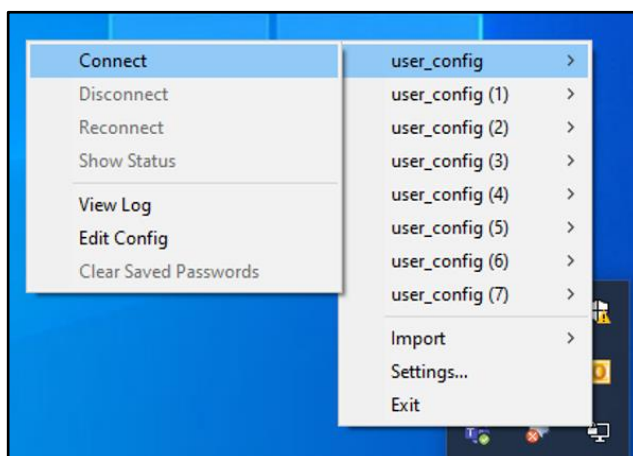


Figure 9. Connect to Config File

5. OpenVPN connection status page appears on the screen.

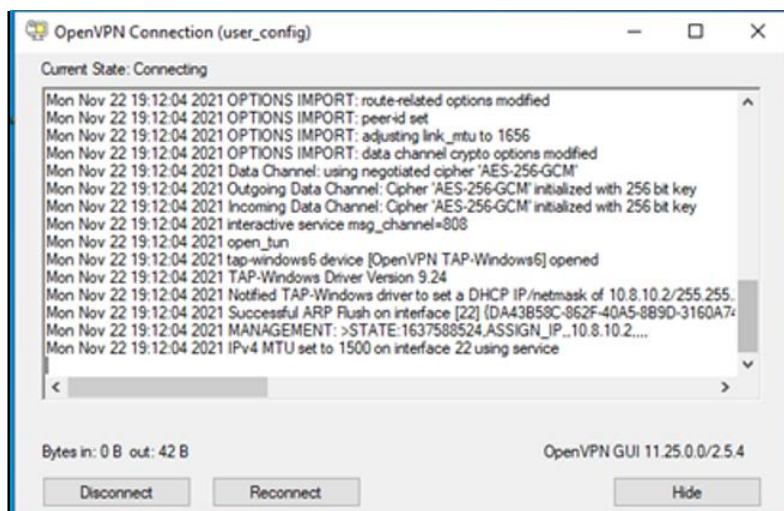


Figure 10. Status of OpenVPN Connection

6. Upon successful tunnel creation, allowed machine modules can be accessed directly using IP address of the module.

6.3.3.1 How to Terminate the Connection

When the work is finished, perform the following steps to terminate the Tunnel VPN connection:

1. Click **Terminate Connection**.
2. As a result, the connection is automatically terminated.



Note: Log messages will be reset if the user logs out and back in again after terminating a connection (**Figure 12**).

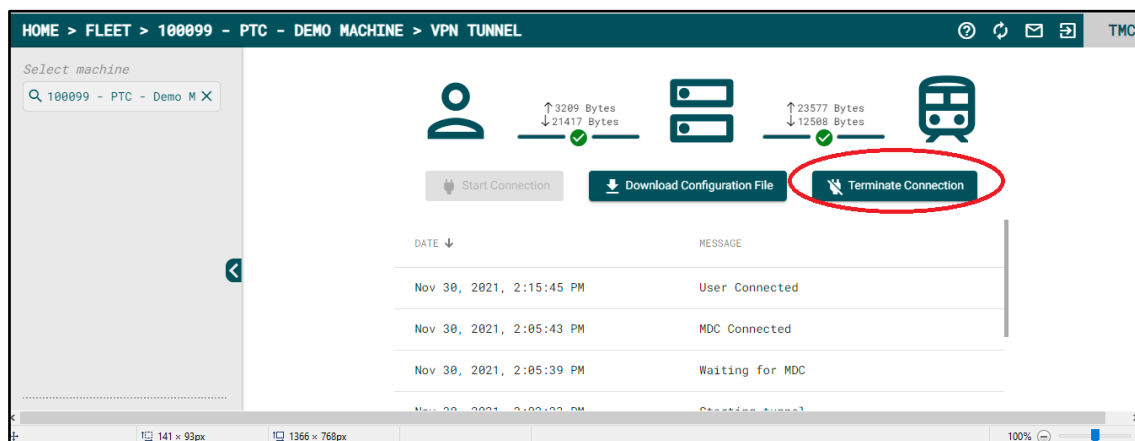


Figure 11. Terminate Connection

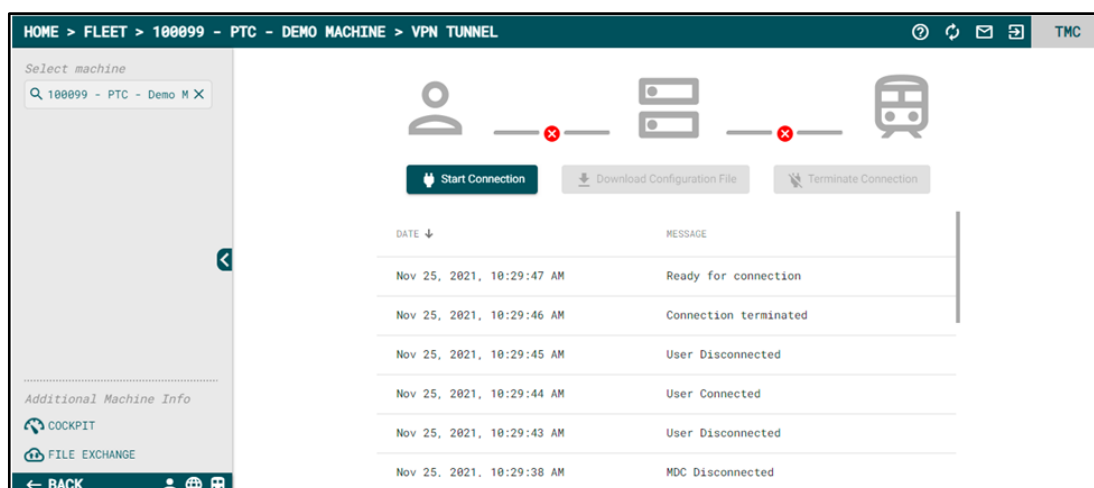


Figure 12. Connection Log Messages

6.3.4 Possible Error Scenarios

The following table provides details of various error status displayed in tmMCO:

Error Status	Issue Description	Workaround / Solution
Terminating connection (VPN not enabled for MDC)	VPN feature is not enabled for the tmMDC.	Send tmMDC a configuration update via MCO registry.
Terminating connection (Two factor authentication failed)	Two factor authentication failed Error	Turn on the VPN switch present with the tmMDC or enable it via PIC.

6.4 VPN Session Continuation (Idle Condition)

- VPN session is terminated if user is idle for 30 minutes.
- Idle time counting starts as soon as "MDC connected" message is shown in tmMCO.
- If user does not connect within 30 minutes, session will be terminated.
- If user connects but do nothing for 30 minutes, session will be terminated.
- User must connect to keep the session running. To keep session running:
 - Linux:** ping <some permitted IP>
 - Windows:** ping -t -l 64 <some permitted IP>
 - Any other traffic from the user to the tmMDC

6.5 User Login / Logout

At first, potential users need to request for user accounts to Administration Manager. After your account is registered, you can log in the system. Refer tmMCO login procedures for logging into the VPN network.

6.5.1 VPN Audit Log

Please refer to **Registry - User Manual**. This guide provides detailed information about the previous VPN connection and VPN log.

7 Administration

The authorized administrator of the **Registry** Server is responsible for integrating the Server into the existing network and providing the Server a high-speed connection to the designated logging device(s) and remote Client workstations.

Please refer to **Registry - User Manual**. This guide provides detailed information about how to create, modify, assign, and delete the users from the Registry application. In addition, this allows the administrator to define VPN group with its access possibility.

8 Troubleshooting

8.1 Maintenance Information

Please refer to **tmMDC - Maintenance Manual**. This guide provides detailed information about how to perform the scheduled preventive maintenance for tmMDC system and its components.

8.2 Error Description

If any defects occur, following error descriptions could be relevant for finding and fixing the defect in a timely manner:

Table 1. IoT Job Status - Error Description

Error Code	Error Description	IOT Job status
0	No error	Succeeded
1	VPN configurations generate error	Failed
2	VPN already running error	Rejected
3	Two factor authentication failed	Rejected
4	Connection to gateway failed	Failed
5	OpenVPN launch error	Failed
6	Configuration parameter missing in IoT job document	Failed
7	VPN feature is not enabled for the tmMDC. And a configuration update is necessary to enable it.	Rejected

8.3 Technical Support and Service

For product support and service, refer to the link below:

<https://tmconnected.com/support>

9 Glossary

Term	Definition
Data Encryption	Encryption is the process of taking the data transmitted through the tunnel established between your VPN client and the server, then scrambling it into an unreadable format known as "ciphertext".
Differential Global Positioning System (DGPS)	A Differential Global Positioning System (DGPS) is an enhancement to the Global Positioning System (GPS) which provides improved location accuracy.
Global System for Mobile Communication (GSM)	GSM is a digital mobile network that is widely used. It makes use of narrowband Time Division Multiple Access (TDMA) technique for transmitting signals. It can also be used for data computing and sending text messages.
Key exchange	This is when the VPN client and server variously encrypt (and decrypt) your data packets using keys or certificates. Basically, they are a series of numbers used to encrypt and decrypt data only they have access to.
Rail Cleaning Machines	A ballast cleaner (also known as an undercutter, a shoulder ballast cleaning machine) is a machine that specialises in cleaning the railway track ballast (gravel, blue stone or other aggregate) of impurities.
Registry	It is a hierarchical database that stores low-level settings for the operating system and for applications that opt to use the registry.
Remote control	Remote control feature allows the user to perform remote operations from tmMCO on a desired tmMDC.
Tamping Machine	A tamping machine is a machine used to tamp the track ballast under railway tracks to make the tracks more durable.
VPN tunnel	VPN tunnel is a private pathway for data to pass through the internet from your computer or mobile device and another network.

